

order of Rate or velocity const:

specific rate or velocity const

specific rate or velocity of rxn.

consumed %	Remaining %	consumed fraction	Remaining fraction
50%	50%	$\frac{1}{2}$	$\frac{1}{2}$
75%	25%	$\frac{3}{4}$	$\frac{1}{4}$
87.5%	12.5%	$\frac{7}{8}$	$\frac{1}{8}$
93.75%	6.25%	$\frac{15}{16}$	$\frac{1}{16}$

$k = \text{specific rate of rxn}$

Unit of k depends upon order of rxn.

order of rxn	Rate law	k	$t_{1/2}$
Zero order	$r = k_0(A)^0$ $r = k_0 = \text{const}$	$k_0 = \frac{x}{t}$ $(A)_t = -k_0 t + (A)_0$	$t_{1/2} = \frac{a}{2k_0}$

① Zero order

$r = k_0(\text{conc})^0$

$M \text{ sec}^{-1} = k_0$

$k_0 = M \text{ sec}^{-1}$
 $\text{mole L}^{-1} \text{ sec}^{-1}$

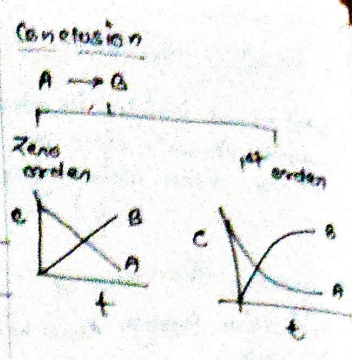
1st order $r = k_1(A)^1$

* Half life of a 1st order rxn is independent of initial concentration.

$k_1 = \frac{2.3 \log \frac{a}{a-x}}{t}$ $t_{1/2} = \frac{0.693}{k_1}$

or $\frac{\ln 2}{k}$

$C_t = C_0 e^{-k_1 t}$
 $A_t = (A_0) e^{-k_1 t}$



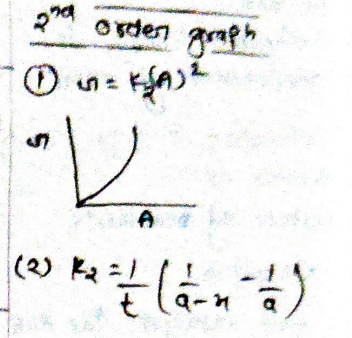
② 1st order

$r = k_1(\text{conc})^1$

$k = \text{sec}^{-1}$
 time^{-1}

2nd order $r = k_2(A)^2$

$k_2 = \frac{1}{t} \left(\frac{1}{a-x} - \frac{1}{a} \right)$ $t_{1/2} = \frac{1}{a k_2}$



③ 2nd order

$r = k_2(\text{conc})^2$

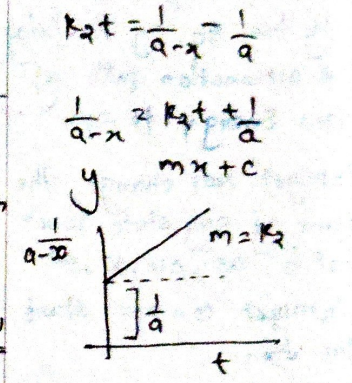
$k_2 = M^{-1} \text{ sec}^{-1}$
 $k_2 = \text{mole}^{-1} \text{ L sec}^{-1}$

nth order $r = k_n(A)^n$

$k_n = \frac{1}{(n-1)t} \left(\frac{1}{(a-x)^{n-1}} - \frac{1}{a^{n-1}} \right)$

$t_{1/2} \propto a^{1-n}$

or $t_{1/2} = \frac{1}{(A)^{n-1}}$



④ nth order

$r = k_n(\text{conc})^n$

$k_n = M^{1-n} \text{ sec}^{-1}$
 $k_n = \text{mole}^{1-n} \text{ L}^{n-1} \text{ sec}^{-1}$

$k = \text{atm}^{1-n} \text{ s}^{-1}$

Graph

Zero order

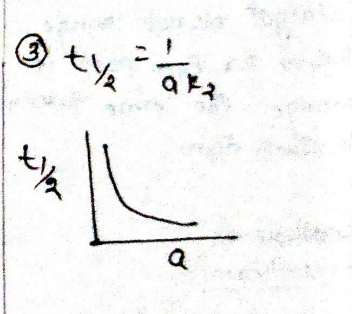
① $r = k_0(A)^0$

② $k_0 = \frac{x}{t}$

③ $(a-x) = -k_0 t + a$

$x = k_0 t$
 $y = m x$
 $m = k_0$

$(a-x) = -k_0 t + a$
 $m = -k_0$



$\log 1 = 0$
 $\log 2 = 0.30$
 $\log 3 = 0.48$
 $\log 4 = 0.60$
 $\log 5 = 0.70$
 $\log 6 = 0.78$
 $\log 7 = 0.84$
 $\log 8 = 0.90$
 $\log 10 = 1$
 $\log 100 = 2$
 $\log 1000 = 3$
 $\log 9 = 0.96$

$\log A^x = x \log A$
 $-\log A = \log \frac{1}{A}$

* Anti log 4.7
10
 $10^4 \times 10^{0.7}$
 5×10^4

$\log A B = \log A + \log B$
 $\log \frac{A}{B} = \log A - \log B$

④ $t_{1/2} = \frac{a}{2k_0}$

$y = m x$

$m = \frac{1}{2k_0}$

* In zero order rxn the concn of A decrease linearly with t & the concn of P increase linearly with t

$A \rightarrow B$

$2NH_3 \rightarrow N_2 + 3H_2$

* $k = \frac{2.3 \log \frac{a}{a-x}}{t_2 - t_1}$

1st order graphs

① $r = k_1(A)^1$
 $y = m x$

② $k_1 = \frac{2.3 \log \frac{a}{a-x}}{t}$

③ $\frac{k_1 t}{2.3} = \log a - \log(a-x)$

$\log \frac{a}{a-x} = \frac{k_1 t}{2.3}$

$\log(a-x) = -\frac{k_1 t}{2.3} + \log a$

$y = -m x + c$
 $m = -\frac{k_1}{2.3}$

④ $C_t = C_0 e^{-k_1 t}$

⑤ $t_{1/2} = \frac{0.693}{k}$

$t_{1/2} = \frac{0.693}{k}$